Environmental Science and Engineering

	Normative 4-year Degree
Major	Environmental Science and Engineering 環境科學及工程學
Degree / Award Title	Bachelor of Engineering in Environmental Science and Engineering 工學士(環境科學及工程學)
Offering Academic Unit	School of Energy and Environment
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	123 / eSTAR:127 / INSPIRE:130 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology	12

	Visit the GE website for more course information.	
School-specified Courses	CA1167 Engineering Communication	3
	SEE1003 Introduction to Sustainable Energy and Environmental Engineering	3
	SEE3002 Energy and Environmental Economics	3
School-specified Courses *(for eSTAR stream)*	CA1167 Engineering Communication	3
	SEE3002 Energy and Environmental Economics	3
	Any course of 3 credit units to be taken in the minor programmes of (i) Computing, (ii) Data Science, (iii) Innovation Management and Entrepreneurship, or (iv) International Business	3

College / School Requirements (18 or 3 credit units)

Catalogue Term : Semester A 2025/26

(i) For taking the major of "Energy Science and Engineering" or "Environmental Science and Engineering" (18 credit units)

Course Code	Course Title	Credit Units
CHEM1200	Discovery in Biology	3
CHEM1300	Principles of General Chemistry	3
PHY1201	General Physics I	3
SEE1000	Professional Development: Career Planning Workshop	0
SEE1002	Introduction to Computing for Energy and Environment	3
SEE2000	Professional Development I	0
SEE4000	Professional Development II	0

Select either MA1200 or MA1300:

Course Code	Course Title	Credit Units
MA1200	Calculus and Basic Linear Algebra I	3
MA1300	Enhanced Calculus and Linear Algebra I	3

Select either MA1201 or MA1301:

Course Code	Course Title	Credit Units
MA1201	Calculus and Basic Linear Algebra II	3
MA1301	Enhanced Calculus and Linear Algebra II	3

(ii) For taking the stream of "Energy/Environment in Science, Technology & Advanced Research (eSTAR)" (3 credits)

Course Code	Course Title	Credit Units
SEE1000	Professional Development: Career Planning Workshop	0
SEE1002	Introduction to Computing for Energy and Environment	3

SEE2000	Professional Development I	0
SEE4000	Professional Development II	0

Major Requirements (no stream/eSTAR:74, INSPIRE:81 credit units)

Catalogue Term : Semester A 2026/27

Notes:

The catalogue term of the major requirement that students will follow will be the effective term of the declared/allocated major.

For normative 4-year degree students who will join the majors allocation exercise, the catalogue term of major requirement will be one year after admission.

For 4-year degree students who already have a major at the time of admission, the catalogue term of major requirement will be the same as their admission term.

1. Basic Core Courses

Normative 4-year Degree / Normative 4-year Degree eSTAR Stream (20 credit units)

Course Code	Course Title	Credit Units
CHEM2004	Principles of Analytical Chemistry	4
MA2181	Mathematical Methods for Engineering	3
SEE2002	Chemical Sciences for Energy and Environmental Engineers	4
SEE2003	Introduction to Energy and Environmental Data Analysis	3
SEE2101	Engineering Thermofluids I	3
SEE2201	Fundamentals of Environmental Engineering	3

Normative 4-year Degree INSPIRE Stream (27 credit units)

Course Code	Course Title	Credit Units	Remarks
CHEM2004	Principles of Analytical Chemistry	4	
MA2181	Mathematical Methods for Engineering	3	
SEE2002	Chemical Sciences for Energy and Environmental Engineers	4	
SEE2003	Introduction to Energy and Environmental Data Analysis	3	
SEE2101	Engineering Thermofluids I	3	
SEE2201	Fundamentals of Environmental Engineering	3	
SEE4993	Research Seminar	1	

SEE4994	Guided Study in Energy and Environment	3	*
SEE4995	Guided Study in Energy and Environment	6	*
SEE4998	Special Project in Energy and Environment	3	*
SEE4999	Special Project in Energy and Environment	6	*

**Remark: Select any one among the following three options:*
- SEE4994 + SEE4998
- SEE4995
- SEE4999

2. Major Core Courses (45 credit units)

Course Code	Course Title	Credit Units
SEE2203	Environmental, Safety, and Occupational Health Management	3
SEE2204	Principles of Sustainability	3
SEE3003	Climate Change and Adaptation Strategies	3
SEE3101	Engineering Thermofluids II	4
SEE3203	Air Pollution	3
SEE4001	Engineers in Society	1
SEE4002	Environmental Engineering Laboratory	3
SEE4004	Environmental Impact Assessment for Sustainable Development	4
SEE4203	Advanced Treatment and Management of Solid and Municipal Waste	3
SEE4204	Environmental Systems Modelling	3
SEE4217	Waste and Wastewater Treatment Engineering	3
SEE4218	Water and Water Resource Engineering	3
SEE4996	Final Year Project	6
SYE4024	Project Management	3

3. Electives (at least 9 credit units)

Select at least **three** courses from the following tables:

Environmental Technology

Course Code	Course Title	Credit Units
CA3677	Hydraulics and Hydrology	3
CHEM4035	Environmental Measurements	4
SEE3207	Indoor Environmental Quality	3
SEE4122	Chemical Separations for Energy and Environmental Applications	3

SEE4216	Combustion and Air Pollution Control	3
SEE4220	Measurements of Air Pollutants	3

Sustainability and Environmental Management

Course Code	Course Title	Credit Units
SDSC3002	Data Mining	3
SEE3001	Energy and Environmental Policy	3
SEE3104	Sustainable and Renewable Energy	3
SEE3204	Urban Sustainability	3
SEE3205	Urban Sustainability	3
SEE3206	Environmental Social Governance	3
SEE4116	Energy and Carbon Auditing	3
SEE4205	Design of Smart Cities and Sustainable Building	3
SEE4206	Social Perspectives of Environmental Science and Engineering	3

Environmental Science

Course Code	Course Title	Credit Units
CHEM4022	Environmental Toxicology	4
CHEM4035	Environmental Measurements	4
CHEM4039	Environmental Conservation and Resources Management	4
SEE3201	Atmospheric Science - An Introductory Survey	3
SEE4202	Atmospheric Chemistry	3
SEE4219	Air Quality Modeling	3

4. Optional Electives (21 credit units)

Students may choose to enroll in all of the following course(s) if they are interested in being a member of The Hong Kong Institution of Engineers (HKIE) in the Building Services (BSS) discipline.

Course Code	Course Title	Credit Units
CA3712	Electrical Services	3
CA3722	HVAC Engineering	3
CA3732	Fire Engineering and Piped Services	3
CA4718	Power Electronics and Smart Lighting Controls	3
CA4737	Fire Science and Modelling	3
SEE2001	Electromagnetic Principles for Energy Engineers	3
SEE3103	Energy Efficiency for Buildings	3

For taking the stream of "Energy/Environment in Science, Technology & Advanced Research (eSTAR)"

The “eSTAR” stream encompasses (i) EVE major; (ii) a research component, and (iii) minor study in one of the four designated areas, namely Computing, Data Science, Innovation Management and Entrepreneurship , and International Business.

(i) EVE major as mentioned above. (74 credit units)

(ii) Research (7 credit units or above)

a. Guided Study and/or Research (Take any one among the following three options for 2 semesters.)

Course Code	Course Title	Credit Units	Remarks
SEE4994	Guided Study in Energy and Environment	3	Option 1
SEE4998	Special Project in Energy and Environment	3	Option 1
SEE4995	Guided Study in Energy and Environment	6	Option 2
SEE4999	Special Project in Energy and Environment	6	Option 3

b. Research Seminar (Students will attend a minimum of 5 research seminars per semester.)

Course Code	Course Title	Credit Units
SEE4993	Research Seminar	1

c. MSc- or PhD- level courses with the approval of the Major Leader. (Optional)

(iii) Minor (15 credit units or above)

Computing (COMP) – 15 credit units

Data Science (DSC) – 18 credit units

Innovation Management and Entrepreneurship (IME) – 15 credit units

International Business (IB) – 15 credit units

Note:
The following courses (11 credit units in total) must be completed before taking DSC minor core courses: CS2311 Computer Programming, MA2503 Linear Algebra, and MA2506 Probability and Statistics.

Suggested Study Plan

Recommended Study Plan of EVE Students